

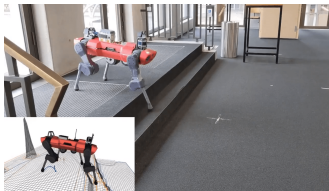
Tutorial on numerical optimal control

Sébastien Kleff

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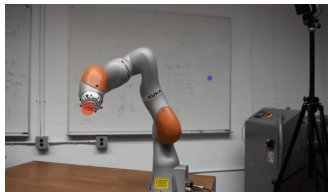
Why optimal control?

Stairs climbing



<https://youtu.be/v6MhPl2ICsc>

Pick-and-place



<https://youtu.be/ZtyCJYsGf4U>

Parkour



<https://youtu.be/tF4DML7FIWk>

Obstacle avoidance



They all solve an **optimal control problem**

Tutorial objectives

Ideally, by the end of this tutorial :

- The relations between OC, MPC & DDP should be clear(er) to you
- You will understand words like "direct multiple shooting"
- You can implement your own MPC to control your favorite robot

To achieve these goals, I will provide

- Quick overview of Optimal Control
- Crash course on nonlinear optimization
- Tutorial using Crocoddyl and mim_solvers

Tutorial (will be) available on a dedicated repo :

https://github.com/skleff1994/mpc_tutorial

Tentative plan :

- Session 1 : Optimal control
- Session 2 : Numerical optimization
- Session 3 : MPC

SESSION 1

Optimal control: Problem definition

Continuous-time **Optimal Control Problem** (OCP)

$$\begin{aligned} V(x_0, t_0) = \min_{u(\cdot)} \int_{t_0}^T \ell(x(t), u(t)) \, dt + \ell_T(x(T)) \\ \text{s.t.} \quad \begin{cases} x(t_0) = x_0 \\ \dot{x}(t) = f(x(t), u(t)) & \forall t \in [t_0, T] \\ c(x(t), u(t)) \leq 0 & \forall t \in [t_0, T] \end{cases} \end{aligned} \quad (1)$$

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Components

- State and control trajectories $x(\cdot) : \mathbb{R}^+ \rightarrow \mathbb{R}^{n_x}, u(\cdot) : \mathbb{R}^+ \rightarrow \mathbb{R}^{n_u}$

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- Initial condition $(x_0, t_0) \in \mathbb{R}^{n_x} \times \mathbb{R}^+$ and horizon $T > 0$ (can be $+\infty$)
- Some technical assumptions

Optimal control: Manipulator example

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A 7-DoF torque-controlled manipulator must reach an end-effector position $p^{\text{des}} \in \mathbb{R}^3$ while using minimum energy and satisfying operating constraints

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- What is x ?
- What is u ?
- What is f ?
- What is ℓ ?
- What is c ?



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A 7-DoF torque-controlled manipulator must reach an end-effector position $p^{\text{des}} \in \mathbb{R}^3$ while using minimum energy and satisfying operating constraints

- We define the state variable $x = (q, \dot{q}) \in \mathbb{R}^{2n_q}$ and the control input $u = \tau \in \mathbb{R}^{n_q}$ with $n_q = 7$
- The dynamics constraint f is given by the robot forward dynamics

$$\underbrace{\begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} \dot{q} \\ M(q)^{-1}(\tau - h(q, \dot{q})) \end{bmatrix}}_{f(x, u)}$$

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A 7-DoF torque-controlled manipulator must reach an end-effector position $p^{\text{des}} \in \mathbb{R}^3$ while using minimum energy and satisfying operating constraints

- Path constraints are e.g. joint limits $[x^{\min}, x^{\max}]$ and torque limits $[u^{\min}, u^{\max}]$

$$c(x, u) = \begin{bmatrix} x - x^{\max} \\ x^{\min} - x \\ u - u^{\max} \\ u^{\min} - u \end{bmatrix}$$

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A 7-DoF torque-controlled manipulator must reach an end-effector position $p^{\text{des}} \in \mathbb{R}^3$ while using minimum energy and satisfying operating constraints

- Initial condition : robot at rest $x_0 = (q_0, 0)$ at $t_0 = 0$
- Cost function : penalizes the distance to the target and the energy

$$\ell(x, u) = \alpha \|u\|_2^2$$

$$\ell_T(x) = \|p(q) - p^{\text{des}}\|_2^2$$

where $\alpha > 0$ is a scalar parameter and the end-effector position $p(q)$ is given by the robot's forward kinematics

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Very generic formulation

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But how to solve this?

Optimal control: Problem definition

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V is a *functional* characterized by the Hamilton-Jacobi-Bellman (HJB) Partial Differential Equation (PDE)

$$-\frac{\partial V(x, t)}{\partial t} = \min_u \left[\ell(x, u) + \mathcal{I}_c(x, u) + \frac{\partial V(x, t)}{\partial x} f(x, u) \right]$$

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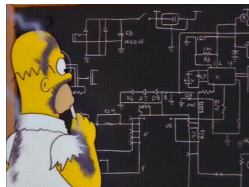
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No analytical solution except in very specific cases !

Analytic solution: Special case of LQR

Particular case : Linear-Quadratic Regulator (LQR)

When f is linear, ℓ is quadratic (no path constraints)

$$V(x_0, t_0) = \min_{u(\cdot)} \int_{t_0}^T \left(x(t)^\top Q x(t) + u(t)^\top R u(t) \right) + x(T)^\top Q_T x(T) \quad (3)$$
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where $Q, Q_T, R \succ 0$ and (A, B) is controllable.

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The value function V is quadratic, i.e. $V(x, t) = x^\top P(t)x$ where $P(t) \succ 0$ is the solution the Riccati differential equation

$$-\dot{P}(t) = A^\top P(t) + P(t)A - P(t)BR^{-1}B^\top P(t) + Q \quad (4)$$

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In the general case, HJB must be solved numerically

Numerical solution: Simple Pendulum

Example : Pendulum swing-up task

State $x = (\theta, \dot{\theta})$, control $u = \tau$ and dynamics model f

$$ml^2\ddot{\theta} + mgl \sin \theta = \tau \quad (5)$$

Running cost $\ell(x, u) = \alpha \|u\|^2$ and terminal cost $\ell_T(x) = \|x\|^2$

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We **discretize** the state and control spaces into finite meshes

We solve the PDE numerically to compute V explicitly for every $(\theta, \dot{\theta})$

Check-out the pendulum example and play with it : `pendulum_bellman.py`

Numerical solution: Simple Pendulum

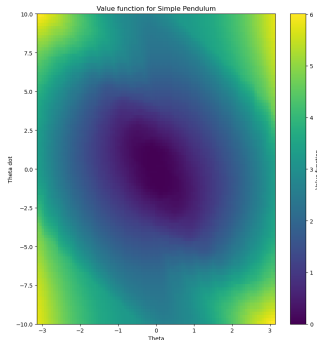
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This is not LQR (f is nonlinear) so we must solve the HJB PDE numerically



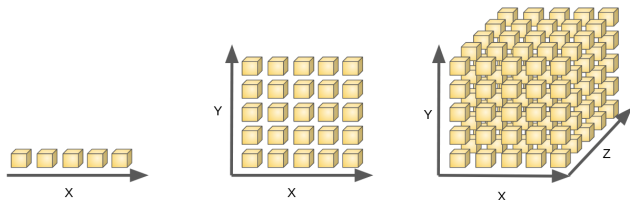
We **discretize** the state and control spaces into finite meshes

We solve the PDE numerically to compute V explicitly for every $(\theta, \dot{\theta})$

Check-out the pendulum example and play with it : `pendulum_bellman.py`

Numerical solution: Curse of dimensionality

Major problem : the number of points N required to maintain the same sampling density increases **exponentially** with the state space dimension n_x



100 points per dimension with $n_x = 2$: $N = 10^4$ points

100 points per dimension with $n_x = 6$: $N = 10^8$ points

Our 7-DoF torque-controlled manipulator has $n_x = 14$...

Computing V explicitly is not tractable if $n_x \geq 4$ or 5

Analytic resolution of HJB PDE : *"Explicit formula for V "*

- Exact *global* solution (i.e. compute $V(x)$ for all x)
- Feedback (closed-loop) policy $\pi(x)$
- Only works in very specific cases (e.g. LQR)

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Luckily there is an alternative : direct optimal control

Direct Optimal Control

- ① "First discretize" : Transform OCP into Nonlinear Program
- ② "Then optimize" : Solve Nonlinear Program

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 - Local solutions only (only valid around some given $x(t)$)
 - Control trajectories $u(t)$ (open-loop policy)*

***Spoiler** : MPC will essentially use direct optimal control to approximate the closed-loop policy $\pi(x)$ through repeated open-loop solutions $u(t)$

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Transcription : Parametrize the infinite-dimensional OCP (2)

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$$x_{k+1} = x_k + \underbrace{F(x_k, u_k)}_{\text{continuous-time dynamics}} \delta \quad (6)$$

$$\ell(x_k, u_k) = \underbrace{L(x_k, u_k)}_{\text{continuous-time cost}} \delta \quad (7)$$

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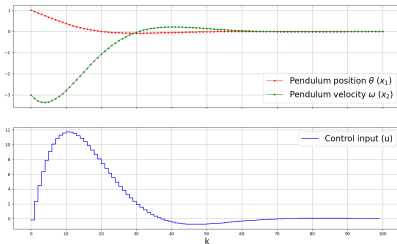
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Pendulum with semi-explicit Euler

$$\dot{\theta}_{k+1} = \dot{\theta}_k + \underbrace{\left(\frac{-g \sin(\theta)}{l} + u \right)}_{\text{continuous-time acceleration}} \delta$$

$$\theta_{k+1} = \theta_k + \delta \dot{\theta}_{k+1}$$



Direct approach: "First discretize"

This leads to a *discrete-time OCP* with *finite-dimensional* decision variables

$$\begin{aligned} \min_{u_0, \dots, u_{T-1}} \quad & \sum_{k=0}^{T-1} \ell(x_k, u_k) + \ell_T(x_T) \\ \text{s.t.} \quad & \begin{cases} x_0 = \hat{x} \\ x_{k+1} = f(x_k, u_k) \\ c(x_k, u_k) \leq 0 \end{cases} \end{aligned} \tag{8}$$

Remarks

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Remarks

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- Most of continuous-time optimal control theory applies in discrete-time
- For instance, the Bellman equation is the discrete-time equivalent of HJB

$$V_j(x) = \min_u V_{j+1}(f(x, u)) + \ell(x, u) + \mathcal{I}_c(x, u) \quad (9)$$

where V_j is the optimal cost-to-go at stage j

Particular case : Linear-Quadratic Regulator (LQR)

When f is linear, ℓ is quadratic (no path constraints)

$$V_0(x_0) = \min_{u_0, \dots, u_{T-1}} \sum_{k=0}^{T-1} \left(x_k^\top Q x_k + u_k^\top R u_k \right) + x_T^\top Q_T x_T \quad (10)$$
$$\text{s.t.} \quad \begin{cases} x_0 = \hat{x} \\ x_{k+1} = A x_k + B u_k \quad \forall k \in \{0, \dots, N-1\} \end{cases}$$

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The value function V is quadratic, i.e. $V_k(x) = x^\top P_k x$ where $P_k \succ 0$ is the solution the Riccati differential equation

$$P_k = A^\top P_{k+1} A - (A^\top P_{k+1} B)(R + B^\top P_{k+1} B)^{-1} B^\top P_{k+1} A + Q \quad (11)$$

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Discrete-time OCP: Special case of LQR

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In the general case, there is no analytic solution for V, π

Direct approach: "Then optimize"

The discrete-time OCP

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Instead, we seek a **local** solution by solving (8) as a Nonlinear Program (NLP)

$$\begin{aligned} \min_z L(z) \\ \text{s.t. } G(z) \leq 0 \end{aligned} \quad (12)$$

where $z \in \mathbb{R}^n$, $L : \mathbb{R}^n \rightarrow \mathbb{R}^+$ and $G : \mathbb{R}^n \rightarrow \mathbb{R}^{n_c}$.

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This is a standard nonlinear optimization problem that can be solved using textbook numerical optimization

- Optimal control is a generic framework
- OCPs are challenging to solve globally
- We can seek local solutions instead
- This requires to transform the original OCP into an NLP

Session 2 will focus on the NLP resolution

- What are the main techniques to solve a generic NLP
- How our NLP has a special structure
- How this structure can be exploited to solve it efficiently

Session 3 will focus on MPC implementation and introduction of the existing tools (Crocoddyl, mim_solvers)